

Docker Swarm

Docker Swarm is Docker's built-in **container orchestration system** — basically, it helps you run and manage **multiple containers across multiple machines** as if they were a single system.

Simple mental model

Think of it like this:

- Without Swarm → you run containers on **one machine**
- With Swarm → you run containers on a **cluster of machines (nodes)**

It handles:

- Deployment
- Scaling
- Load balancing
- Failover (self-healing)

You take multiple servers and turn them into one **cluster**, then deploy your app to that cluster.

Key Concepts (super important)

1. Node

A machine in the cluster

- **Manager Node**
 - Controls everything
 - Schedules containers
- **Worker Node**
 - Runs your containers

2. Service

This is the main abstraction.

Instead of:

docker run ...

You do:

docker service create ...

A service = desired state

- “I want 3 instances of my app running”

3. Tasks / Containers

Each service runs containers called **tasks**

Example:

- Service → 3 replicas
- Tasks → 3 containers

4. Scaling

docker service scale my-app=5

Swarm will:

- Automatically create 5 containers
- Distribute them across nodes

5. Load Balancing

Swarm automatically:

- Distributes traffic across containers
- No need for manual setup

6. Self-Healing

If a container crashes:

- Swarm **restarts it automatically**

If a node dies:

- Swarm **moves containers to another node**

Example Flow (important for understanding)

1. Initialize swarm:

`docker swarm init`

2. Add workers:

`docker swarm join ...`

3. Deploy app:

`docker service create --replicas 3 nginx`

4. Swarm ensures:

- 3 containers always running
- Spread across nodes

Why it exists

Without orchestration:

- You manually manage containers ❌
- No scaling ❌
- No failover ❌

With Swarm:

- Automated infrastructure ✅